

INDUSTRIAL TRAINING PROJECT

Artificial Intelligence — Project 1

RULE-BASED AI CHATBOT ■

Completed Internship Task / Project Report

Batch: 2026

Powered by: DecodeLabs

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1. Project Overview

The supplied DecodeLabs training kit defines Project 1 as a foundation exercise in Artificial Intelligence: building a Rule-Based AI Chatbot. The project focuses on control flow and logic rather than deep learning. The chatbot simulates basic human interaction by following explicit if-else instructions.

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The stated goal is to create a simple rule-based chatbot that responds to predefined user inputs. The required behavior includes handling greetings and exit commands, using if-else logic for responses, and running continuously in a loop. ■filecite■turn0file0■L31-L45■

2. Objectives

- Understand basic rule-based AI concepts.
- Apply if-else decision-making.
- Create a continuous input-response loop.
- Handle common greetings and exit commands.
- Produce a working command-line chatbot that demonstrates the project requirements.

3. Tools and Technologies

- Programming language: Python 3
- Development environment: VS Code, IDLE, or Google Colab
- Core concepts: variables, input/output, strings, conditional statements, while loop, break statement.

4. System Design

The chatbot follows a simple cycle: receive user input → normalize the input → check predefined rules → generate a response → repeat until an exit command is received.

Flow:

```
START | v Ask user for input | v Convert input  
to lowercase | v Is input an exit command? ---- YES  
----> Goodbye -> STOP  
|  
NO | v Is it a greeting? ----- YES ----->  
Greeting response  
|  
NO | v Is it another known command? -> YES ---->  
Appropriate response  
|  
NO | v  
Default response  
| v  
Repeat loop
```

5. Complete Python Implementation

The following implementation directly satisfies the supplied project requirements: predefined responses, greeting and exit handling, if-else decision logic, and a continuous loop.

```
# DecodeLabs - Project 1
# Rule-Based AI Chatbot

def chatbot():
    print("=====")
    print("  Welcome to the Rule-Based AI Bot")
    print("=====")
    print("Type 'bye', 'exit', or 'quit' to stop.")
    print()

    while True:
        user_input = input("You: ")
        user_input = user_input.strip().lower()

        # Exit commands
        if user_input in ["bye", "exit", "quit"]:
            print("Bot: Goodbye! Have a great day!")
            break

        # Greetings
        elif user_input in ["hi", "hello", "hey", "good morning", "afternoon", "good evening"]:
            print("Bot: Hello! How can I help you?")

        # Basic predefined questions
        elif user_input in ["how are you", "how are you?"]:
            print("Bot: I am doing great! Thanks for asking.")

        elif user_input in ["what is your name", "what is your name?"]:
            print("Bot: I am a simple Rule-Based AI Chatbot.")

        elif user_input in ["what can you do", "what can you do?"]:
            print("Bot: I can respond to predefined messages using rules.")

        elif user_input in ["help", "help me"]:
            print("Bot: Try saying hello, asking my name, or asking what I can do.")

        # Default response
        else:
            print("Bot: Sorry, I don't understand that yet.")

if __name__ == "__main__":
    chatbot()
```

6. Explanation of the Code

chatbot(): Contains the complete chatbot program. **while True:** Keeps the chatbot running continuously. **input():** Accepts a message from the user. **strip().lower():** Removes extra spaces and converts the message to lowercase so matching is easier. **if / elif / else:** Implements the rule-based decision-making system. **break:** Ends the loop when the user enters an exit command.

7. Sample Test Cases

User Input	Expected Bot Response
hello	Hello! How can I help you?
hi	Hello! How can I help you?
how are you?	I am doing great! Thanks for asking.
what is your name?	I am a simple Rule-Based AI Chatbot.

User Input	Expected Bot Response
what can you do?	I can respond to predefined messages using rules.

help	Try saying hello, asking my name, or asking what I can do.
xyz	Sorry, I don't understand that yet.
bye	Goodbye! Have a great day!

8. Expected Output

```
=====
Welcome to the Rule-Based AI Bot
=====
Type 'bye', 'exit', or 'quit' to stop.

You: hello Bot: Hello! How can
I help you?

You: what is your name?
Bot: I am a simple Rule-Based AI Chatbot.

You: what can you do?
Bot: I can respond to predefined messages using rules.

You: something unknown Bot: Sorry, I
don't understand that yet.

You: bye Bot: Goodbye! Have a
great day!
```

9. Testing and Verification

The implementation can be verified by running it and entering each test input from the table. Successful completion means the chatbot keeps accepting messages, returns the appropriate predefined response for supported inputs, gives a default response for unsupported inputs, and terminates when an exit command is entered.

10. Learning Outcomes

After completing this project, the intern demonstrates practical understanding of control flow, decision-making logic, predefined rules, and continuous program execution. These skills correspond directly to the training kit's stated key skills: control flow, decision-making logic, and basic AI concepts.

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11. Possible Enhancements

The training kit encourages experimentation, including expanding the bot's vocabulary, adding nested conditions for smarter responses, and giving the chatbot a unique personality. ■filecite■turn0file0■L47-L62■

Possible next steps include:

- Add more greetings and conversational rules.
- Add more predefined questions and answers.
- Organize rules into functions for cleaner code.
- Add simple keyword matching.
- Give the chatbot a customized personality while keeping the rule-based design.

12. Conclusion

Project 1 successfully implements a basic Rule-Based AI Chatbot using Python control flow. The solution follows the supplied DecodeLabs brief by using explicit if-else rules, handling greetings and exit commands, and operating inside a continuous loop. The training kit presents this project as the essential starting point for interns and emphasizes hands-on practice as the route to mastering foundational AI concepts.

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